

Reading Calibrated Uncertainty from Language Model Trajectories

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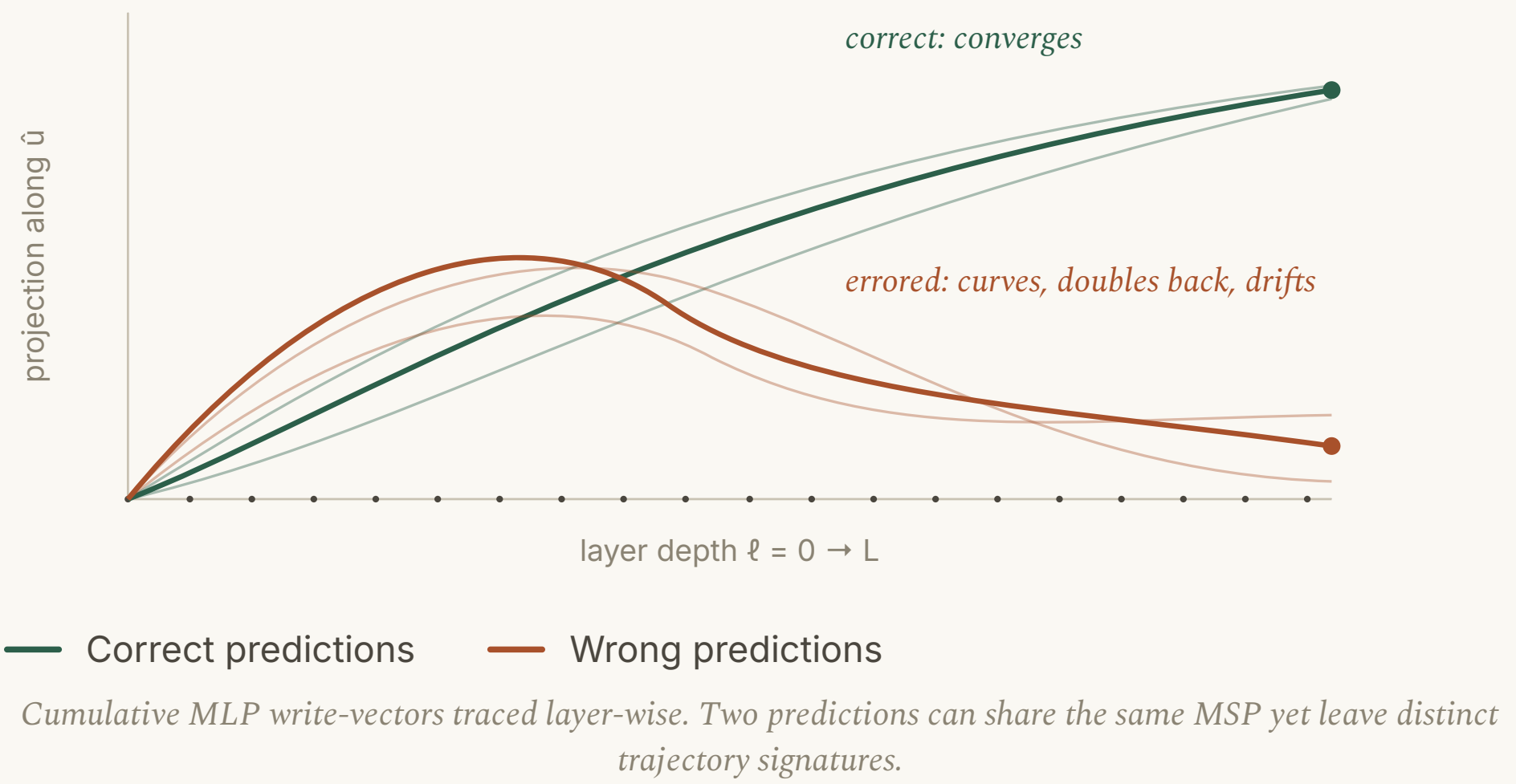
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Read activations as **motion**, not isolated states. How a representation moves through depth reveals uncertainty its final probability hides.

I The problem with MSP

The default uncertainty signal, **Maximum Softmax Probability**, incurs no additional computational cost but is often **miscalibrated**, assigning high confidence even when the answer is wrong. Prior probing reads hidden states *in isolation*, discarding the layer-wise path that formed them.

Similar endpoints arise from very different paths. **How evidence accumulates, reinforces, or reverses across depth** reveals what final probabilities collapse.



II Method

- Capture the trajectory**
One forward pass. Hook each block's MLP write-vector m_ℓ at the answer position; partial sums $s_\ell = \sum m_k$ trace a path through residual-stream space; its endpoint is s_L , with unit direction $\hat{u}$.
- Eleven scale-invariant features**
Closed-form geometric descriptors of the path, comparable across hidden dimensions and model families.
- Sparse linear probe**
Elastic-net logistic regression on the features and their MSP interactions, interpretable end to end.

III What the geometry measures

	FORMULA	WHAT IT CAPTURES
<i>Depth allocation</i>		
Relative update magnitude	$\ m_\ell\ / \bar{n}$	Per-layer update size vs. the mean
Cumulative path fraction	$T^{-1} \sum \ m_k\ $	Front- vs. back-loading of computation
<i>Local shape</i>		
Consecutive cosine	$\cos(m_{\ell-1}, m_\ell)$	Directional consistency of updates
Curvature	$1 - \cos(m_{\ell-1}, m_\ell)$	Abrupt changes of direction
Update-state alignment	$\cos(m_\ell, s_{\ell-1})$	Reinforces vs. contradicts prior state
<i>Endpoint alignment</i>		
Direction to final	$\cos(s_\ell, s_L)$	State convergence to the endpoint
Update to final	$\cos(m_\ell, s_L)$	Update alignment with endpoint direction
Signed final support	$\langle m_\ell, \hat{u} \rangle / \ m_\ell\ $	Same quantity, inner-product framing
Contradictory support	$\max(0, -\langle m_\ell, \hat{u} \rangle / \ m_\ell\)$	Magnitude of opposition to the endpoint
Orthogonal mass fraction	$\sqrt{1 - \cos^2(m_\ell, s_L)}$	Update share orthogonal to the endpoint
<i>Trajectory efficiency</i>		
Cumulative coherence	$\ s_\ell\ / \sum \ m_k\ $	Net displacement vs. path length

Expect high uncertainty on errors and low on correct answers, even the errors **MSP is confident about. Abstain, defer, or escalate accordingly.**

Scope: discrete-choice settings; probe fit per (model, task). Cross-task transfer remains open.

9 LLMs • 3 families • 3B to 72B • Qwen • Llama • DeepSeek
github.com/s2labres/uq-trajectory

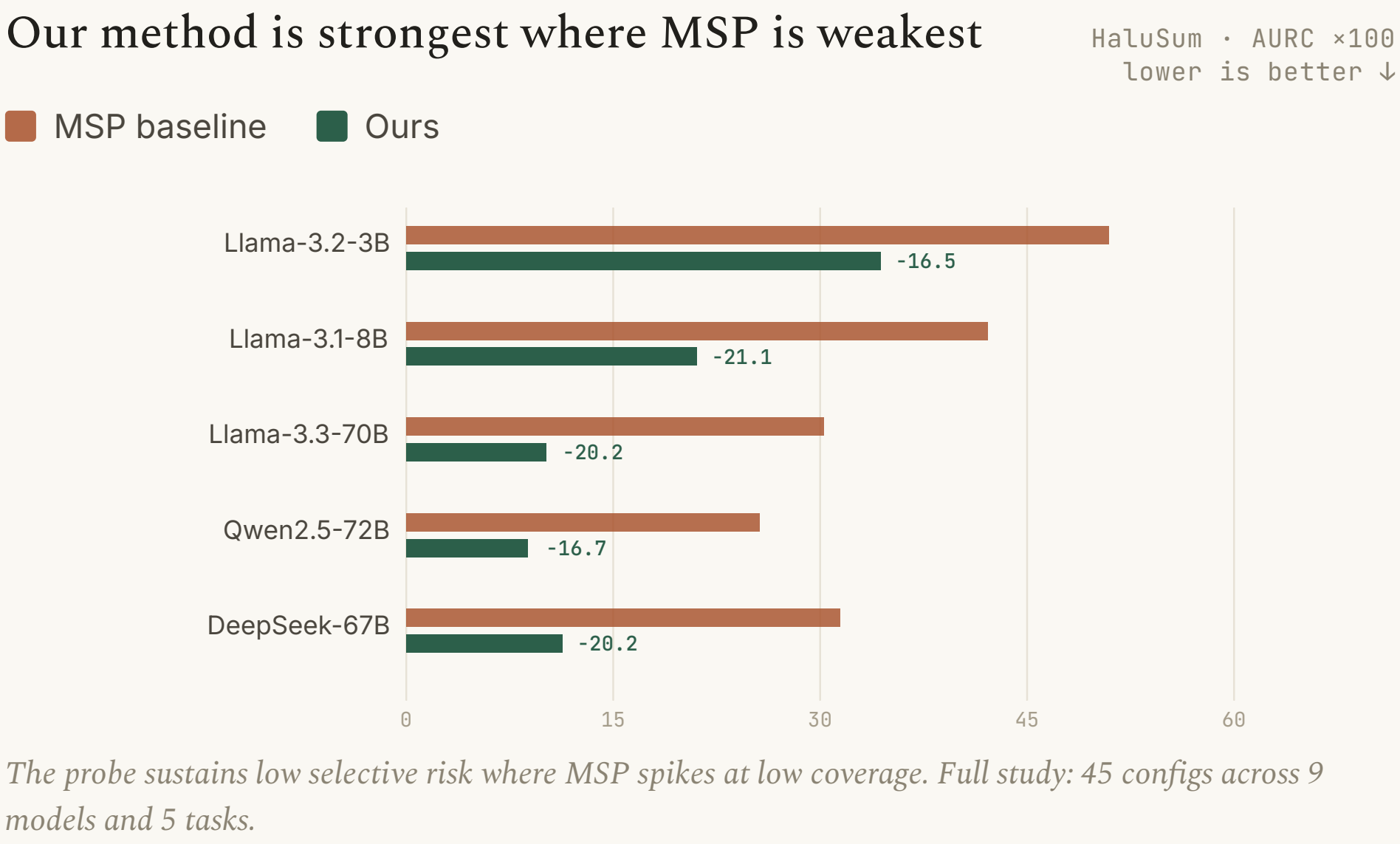
IV Results

41/45

model-dataset pairs improved over MSP

Up to **21.8 AURC points** recovered. Gains scale with baseline miscalibration (**Spearman $\rho = 0.78$**), largest exactly where MSP fails worst.

AURC Area Under the Risk-Coverage curve averages the error rate of retained predictions as the confidence threshold descends from most to least certain. A lower score indicates a confidence signal that reliably concentrates errors among abstained samples.



V Key findings

- Independent.** Trajectory features alone beat MSP on 35/45 configs.
- Complementary.** Trajectory features with MSP lift to 41/45 configs.
- Discriminative.** At fixed MSP ≈ 0.97 , correct and incorrect predictions leave distinct trajectory signatures.

VI Failures become legible

Because every coefficient has a geometric meaning, the probe reveals not only **whether** a model will err but **how**: which layers commit prematurely, which contradict the running state, and where trajectories drift from their endpoint. The signatures are task-specific.

